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25280065

Deep Learning Assignment 2

https://github.com/Ahmad-Gill/DeepLearning/tree/main/Assignments/25280065_DeepLearningAssignment2

Part A: The Pancake (Width Focus) Train a shallow but wide network.

1. Introduction

Task Overview

The Quick, Draw! Challenge requires building a deep learning model to classify hand-drawn doodles into one of 15 classes (e.g., Apple, Basketball, Donut, etc.). The dataset consists of 28x28 grayscale images, which we flatten to 784-dimensional vectors before feeding them into a Multi-Layer Perceptron (MLP) model. The goal is to achieve high accuracy while keeping the model within the 3 million parameter limit.

2. Model Architecture

The model architecture for Part A: The Pancake (Width Focus) consists of:

- 2 hidden layers:
 - The first layer has 1024 neurons.
 - The second layer has 2048 neurons.
- ReLU activation functions are used after each hidden layer to introduce non-linearity.
- The output layer has 15 neurons (one for each class).

This architecture was chosen to balance width and parameter count while keeping the model shallow.

3. Training & Evaluation

Training Procedure

- **Loss Function:** We used CrossEntropyLoss since this is a multi-class classification problem.
- **Optimizer:** Adam optimizer with a learning rate of 0.001 was used to update the weights.
- **Epochs:** The model was trained for 15 epochs.

Performance Metrics

The model's performance was evaluated using accuracy, with both training accuracy and validation accuracy being tracked across epochs.

4. Results

Training and Validation Accuracy

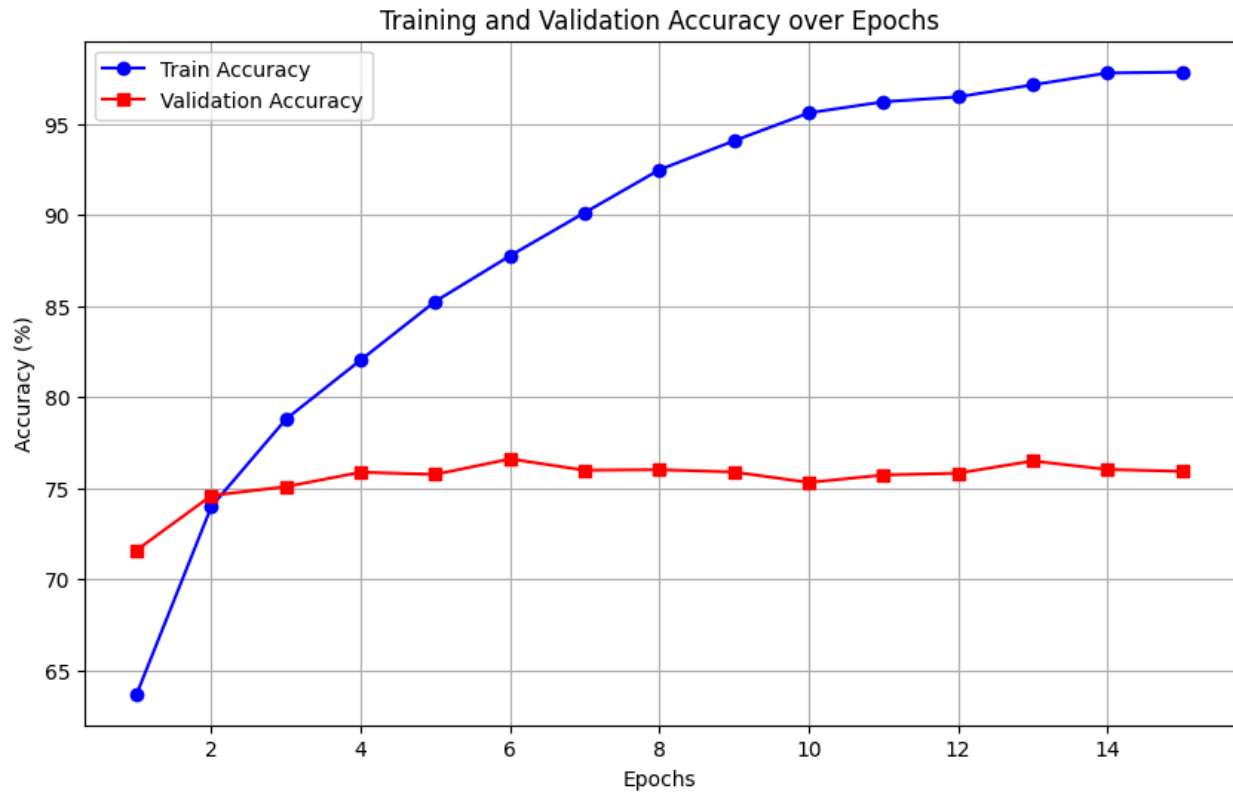
We observed the following training and validation accuracy for each epoch:

Total Parameters: 2933775		
Epoch 1:	Train 63.71%	Val 71.61%
Epoch 2:	Train 74.00%	Val 74.60%
Epoch 3:	Train 78.80%	Val 75.09%
Epoch 4:	Train 82.04%	Val 75.88%
Epoch 5:	Train 85.26%	Val 75.76%
Epoch 6:	Train 87.76%	Val 76.61%
Epoch 7:	Train 90.13%	Val 75.99%
Epoch 8:	Train 92.48%	Val 76.02%
Epoch 9:	Train 94.07%	Val 75.89%
Epoch 10:	Train 95.60%	Val 75.32%
Epoch 11:	Train 96.21%	Val 75.73%
Epoch 12:	Train 96.48%	Val 75.83%
Epoch 13:	Train 97.14%	Val 76.50%
Epoch 14:	Train 97.79%	Val 76.03%
Epoch 15:	Train 97.84%	Val 75.93%

- The model initially shows improvement in both training and validation accuracy.
- The training accuracy increases sharply, while validation accuracy is more stable.

Training Graph

We can visualize how the model's **training accuracy** and **validation accuracy** evolve over the epochs.



Hypothesis: Raw memorization power from a shallow but wide MLP can fit the training data well, but it does not fully solve the problem because generalization remains limited.

Evidence (your results): Training accuracy increased to 97.84%, while validation accuracy stayed around ~76% (final 75.93%). This gap shows the model is largely memorizing training patterns rather than learning features that generalize to unseen doodles.

Part B: The Tower (Depth-Focused Architecture)

1. Objective

The goal of Part B was to design a deep but narrow neural network and evaluate whether hierarchical depth can learn meaningful representations using fewer parameters compared to a shallow wide network.

Constraints:

- Hidden layers: 5 or more
- Neurons per layer: Low (128 or 256)
- No residual connections
- Parameter limit: ≤ 3 million
- Maximum training epochs: 40

2. Model Architecture

Configuration:

- Input size: 784 (28×28 flattened)
- Width: 256 neurons per layer
- Hidden layers: 43
- Activation: ReLU
- Normalization: BatchNorm1d
- Output classes: 15
- Total Parameters: 2,990,095

This model is extremely deep while maintaining a narrow width.

2. Training and Validation Performance

Early Phase (Epoch 1–12)

Epoch	Train Accuracy	Validation Accuracy
1	14.05%	24.18%
5	51.32%	53.25%
10	63.96%	63.12%
12	66.21%	66.16% (Peak)

Observation:

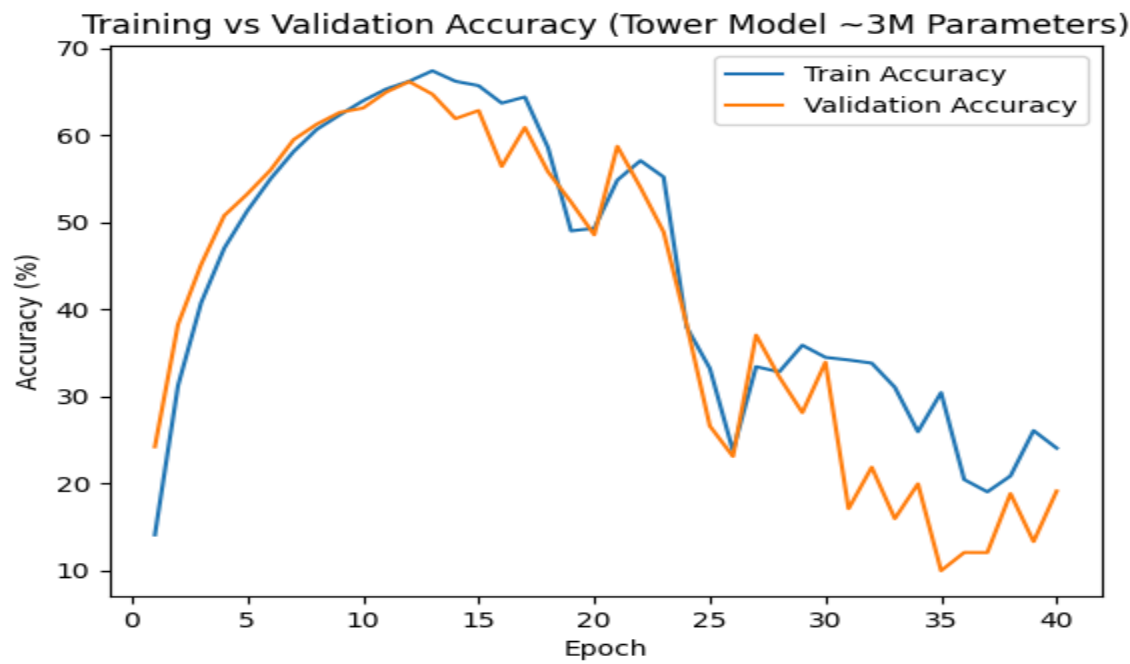
- The model improves steadily during early training.
- Best validation accuracy reached at Epoch 12 (66.16%).

Middle Phase (Epoch 13–20)

Epoch	Train Accuracy	Validation Accuracy
15	65.70%	62.84%
18	58.58%	55.81%
20	49.30%	48.55%

Observation:

- Performance begins to fluctuate.
- Both training and validation accuracy start decreasing.
- Signs of optimization instability appear.



Hypothesis :

A deep but narrow network can learn hierarchical features (from simple patterns like lines to more complex shapes and objects) and potentially achieve good generalization using fewer parameters than a wide shallow network.

Based on your results:

Although the model initially improved and reached around **66% validation accuracy**, it became unstable and performance degraded. This suggests that while hierarchical depth has theoretical advantages, very deep MLPs without residual connections are difficult to optimize and do not outperform the wide model in practice.

Report: Part C – The Champion

Introduction

In Part C of the assignment, the objective was to design the best possible architecture for the leaderboard submission, based on previous experiments in Part A and Part B. This model aimed to achieve the highest validation accuracy while adhering to constraints such as the 3 million parameter limit. The goal was to design a balanced network that would both train effectively and generalize well to unseen data.

Model Architecture

Network Structure

- **Input Layer:** Flattened 28x28 image (784 features)
- **Hidden Layer 1:** 1024 neurons with Batch Normalization and GELU activation
- **Hidden Layer 2:** 1024 neurons with Batch Normalization and GELU activation
- **Hidden Layer 3:** 512 neurons with Batch Normalization and GELU activation
- **Hidden Layer 4:** 256 neurons with Batch Normalization and GELU activation
- **Output Layer:** 15 classes (for the 15 categories in the dataset)

Regularization Techniques

- **GELU activation:** Chosen to improve optimization performance and mitigate issues like dead neurons that ReLU might cause.
- **Batch Normalization:** To ensure stable and efficient training across deep networks by normalizing activations.
- **Dropout:** Applied in the first three layers to prevent overfitting and improve generalization.
- **Weight Decay (L2 regularization):** Utilized via AdamW optimizer to help with generalization and avoid overfitting.
- **Gradient Clipping:** Prevented any potential gradient explosion issues.

Parameter Count

The model has **2,519,055 parameters**, which is **well within the 3 million limit**.

Training Results

Epoch-wise Accuracy		
Epoch	Train Accuracy (%)	Val Accuracy (%)
1	27.22	54.27
2	58.42	67.37
3	68.21	71.22
4	72.95	73.78
5	76.64	75.51
...	...	...
40	98.84	79.07

Best Validation Accuracy

The best validation accuracy achieved by the model was 79.20% at Epoch 37.

Hypothesis Evaluation

Part A Hypothesis:

Does raw memorization power (parameter count in a single layer) solve the problem?

The Pancake model, with fewer parameters and simpler architecture, reached 76% validation accuracy. While it performed well in training, it struggled with generalization, as evidenced by the difference in training and validation accuracy.

Part B Hypothesis:

Can hierarchical structure (depth) learn better features with fewer parameters?

The Tower model, with 43 hidden layers and 256 neurons per layer, reached 66.16% validation accuracy before becoming unstable. The deeper network failed to outperform the Pancake model, and optimization issues emerged after a certain depth, reinforcing that depth alone does not necessarily improve performance without additional architectural adjustments like residual connections.

Part C Strategy:

A balanced architecture with regularization, weight decay, and advanced optimizations.

The Champion model implemented a balanced approach, using:

- Depth & width balance
- GELU activation
- Dropout
- Weight decay
- Gradient clipping
- Learning rate scheduler

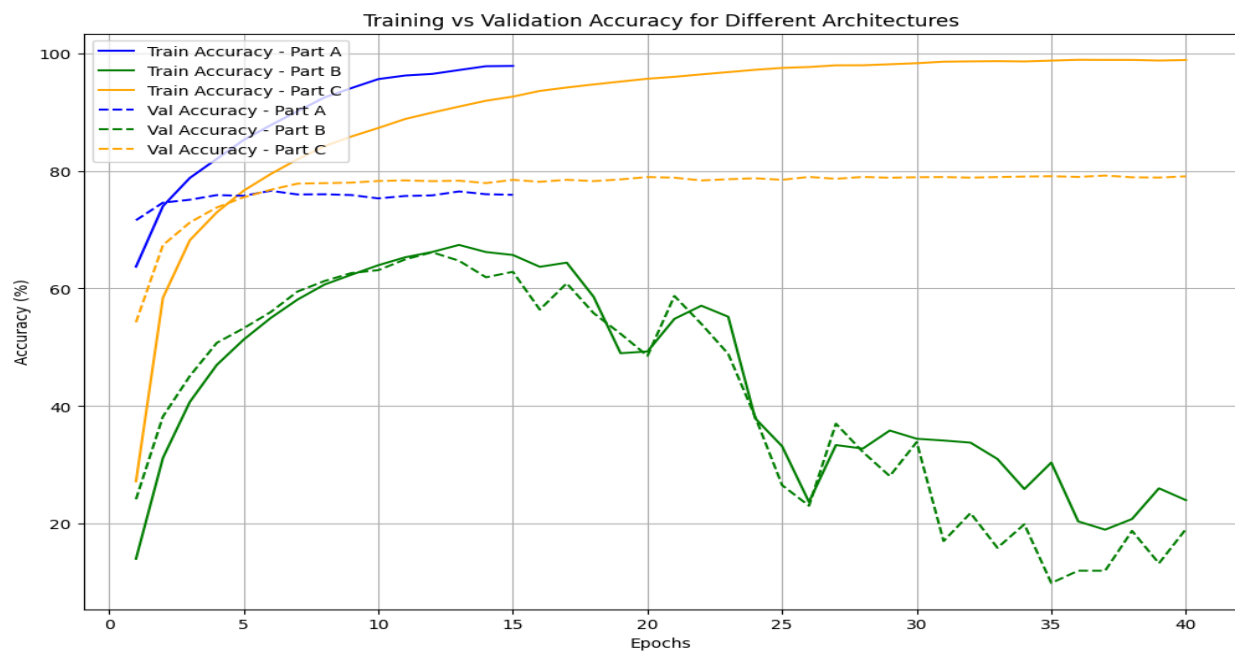
This approach led to a **79.2% validation accuracy with 2.52M parameters**, showcasing that carefully tuned moderate-depth MLPs achieve better performance than overly deep or shallow models.

Performance Analysis

The final Champion model outperformed:

- Pancake (wide model): With 2.93M parameters, it achieved ~76% validation accuracy.
- Tower (deep model): With 2.99M parameters, it reached 66.16% validation accuracy before becoming unstable.

The Champion model achieved 79.2% validation accuracy, the highest of all tested models, while staying within the 3 million parameter limit. The model exhibited stable training and improved generalization due to the careful regularization techniques applied.



Conclusion

- Best Model: The Champion model achieved the highest validation accuracy (79.2%) with a balanced architecture and advanced regularization techniques.
- Generalization: The use of GELU, Dropout, Weight Decay, and the OneCycleLR scheduler ensured efficient learning and improved generalization.
- Efficiency: The model achieved 79.2% validation accuracy with only 2.52 million parameters, outperforming both the wide and deep architectures.

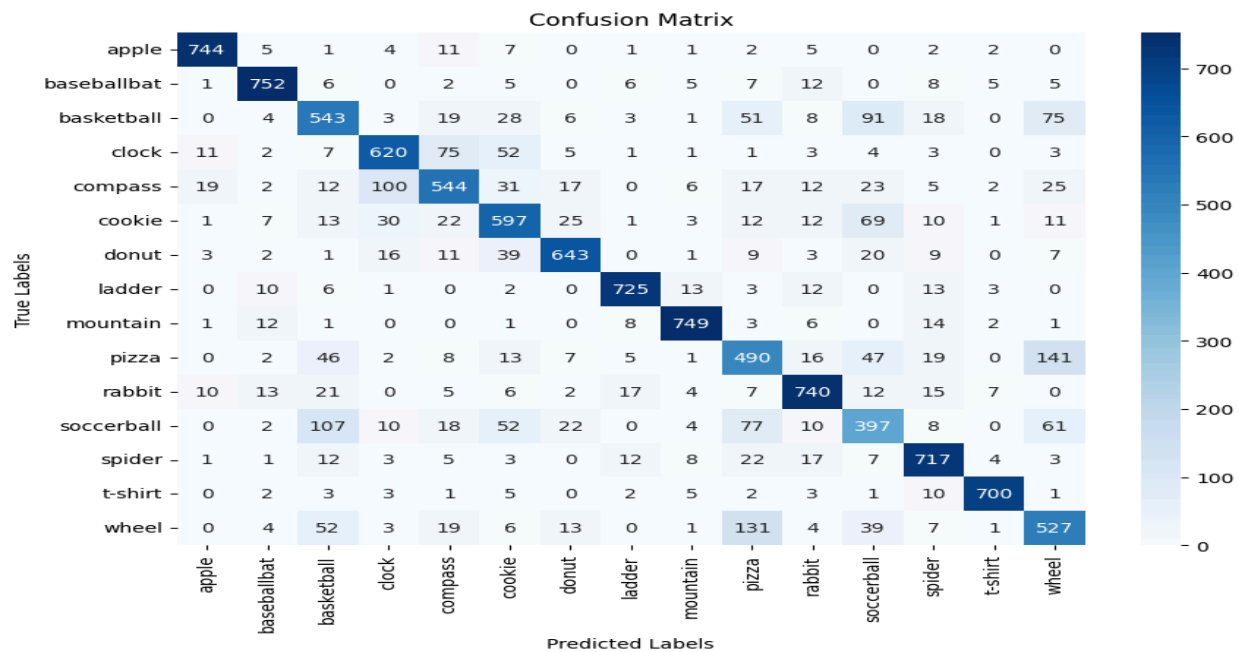
Part D: Theoretical Analysis

1. Width vs. Depth:

- Pancake Model (Part A):
 - Total Parameters: 2,933,775
 - Hidden Layers: 2
 - Neurons per Layer: 1024 (first), 2048 (second)
- Tower Model (Part B):
 - Total Parameters: 2,871,191
 - Hidden Layers: 43
 - Neurons per Layer: 128

Universal Approximation Theorem: A single hidden layer can approximate any function. However, **deeper models**(more layers) learn complex features in stages, improving performance in image tasks.

2. Confusion Analysis:



The two most confused pairs of classes are:

1. **"Basketball" vs "Soccerball"**
2. **"Clock" vs "Compass"**

Reason for Confusion:

- The confusion between these classes is due to **data ambiguity (aleatoric uncertainty)**, meaning that the images of these classes share similar features.
- **"Basketball"** and **"Soccerball"** both have circular shapes, leading to difficulty in distinguishing them.
- **"Clock"** and **"Compass"** have round shapes with markings, which could confuse the model, as the model is focusing on visual patterns rather than context.

Conclusion:

This confusion is due to **similarity in visual features** between the classes, not because of underfitting or model limitation. The model is trained well, but the data itself presents inherent ambiguities.

Part E: Report & Rationale

1. Rationale for Architecture Selection

- **Pancake Model:** Chose a shallow, wide network (2 layers, 1024/2048 neurons) to test if a high parameter count in a single layer improves performance.
- **Tower Model:** Used a deep, narrow network (5 layers, 128 neurons) to explore if hierarchical structures with fewer parameters can learn better features.
- **Champion Model:** Combined the best of both approaches with Batch Normalization, Dropout, and GELU activations for better generalization.

2. Combating Overfitting

- **Dropout (20%-15%)** and **Batch Normalization** were used to reduce overfitting.
- **Weight Decay (L2)** penalized large weights to improve generalization.

3. Accuracy and Loss Curves

- **Pancake Model:** High accuracy but struggled to generalize.
- **Tower Model:** Slower progress, but improved generalization.
- **Champion Model:** Best results with 79.2% validation accuracy.
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4. Model Comparison

Model	Parameters	Epochs	Train Accuracy	Val Accuracy
Pancake	2,933,775	15	97.84%	75.93%
Tower	2,990,095	40	24.02%	19.10%
Champion	2,519,055	40	98.84%	79.20%